

Yanghao Chen

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Education

Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

Research Interests: Solid Mechanics, Scientific Machine Learning, Neural Operators.

Research Projects

GPU-Accelerated Hybrid FEM–Neural Operator Coupling Framework

Jan. 2026 – Present

Lead Undergraduate Researcher; Advisor: [Prof. Somdatta Goswami](#)

Johns Hopkins University

- Generated displacement and stress data using GPU-accelerated JAX-FEM simulations to train a local neural operator.
- Trained Transolver in JAX to learn mappings from interface displacement fields to stress fields within the local subdomain.
- Computed interface reaction forces from predicted stresses and coupled the neural-operator subdomain with the outer FEM domain through iterative displacement–force exchange.

PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher; Advisor: [Prof. Xianyang \(Tom\) Chen](#)

Tongji University

- Trained a PINN in PyTorch to predict shock propagation in the one-dimensional inviscid Burgers equation, using PDE-residual, initial-condition, and boundary-condition losses.
- Added constant artificial viscosity to the PDE residual, using automatic differentiation to compute the required derivatives.
- Compared predicted shock profiles with the analytical inviscid solution, examining transition width and local deviations near the discontinuity.

ConvLSTM Modeling of Chemo-Mechanical Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Automated polycrystalline NMC microstructure generation and MATLAB–COMSOL simulations to produce time-aligned images of lithium concentration and von Mises stress.
- Trained separate conditional ConvLSTM models in PyTorch to predict future sequences of concentration and stress fields from past field images, grain-orientation maps, and C-rate inputs.
- Used MSE and SSIM losses with scheduled sampling and patch-based training for multi-step autoregressive prediction.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Curated instruction–response datasets from textbooks and technical literature on stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B using LoRA for question answering and concept explanation in mechanics of materials.
- Released two instruction datasets and two Qwen2.5-based model checkpoints on [Hugging Face](#).

Technical Skills

Scientific Machine Learning

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM

Computational Mechanics

Finite Element Methods, FEniCSx, Gmsh, COMSOL Multiphysics, ABAQUS

Scientific Computing & Tools

Python, MATLAB, Linux, Git/GitHub, LiveLink for MATLAB

LLM Fine-Tuning

Qwen2.5, LoRA/PEFT, Instruction Dataset Curation

Languages

Chinese (native), Cantonese, English